Detection of Brain Tumours Using Supervised Learning Models

Aidan Mistry (46510)

BAA1027

Doctor Michael Farayola

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Abstract

Brain tumour detection is a difficult and critical medical diagnosis due to the complexity and unpredictability of tumour structures, and the high risk of not detecting or delaying the identification of a tumour. This study investigates how effective machine learning (ML) algorithms are at identifying brain tumours. A dataset containing 3,762 instances with static and texture-based features was analysed after being processed and standardised. (Bohaju, 2020) The paper utilises three supervised learning models, K-Nearest Neighbour (KNN), Decision Tree (DT), and Random Forest (RF), which were used and evaluated with accuracy and confusion matrices.

The results show that all three models had a high rate of identification, with all three models exceeding 97% accuracy. Both KNN and Random Forest performed slightly better, achieving an accuracy of 98.4%, while the decision tree performed slightly worse. The accuracies indicate that feature-based ML approaches are effective for detecting tumours. The study concludes that traditional ML models are capable of achieving high accuracy in tumour detection when data is correctly formatted and preprocessed. The results suggest that KNN and Random Forest are equal, but since Random Forest avoids overfitting, it would likely be the model most suitable for proper medical application.

Introduction

Brain tumours represent a significant global health challenge due to high mortality rates and their complexity to diagnose. They arise from uncontrolled cell growth within the brain, often leading to severe neurological damage and, in many cases, death if not detected early (Ucuzal et al., 2019). According to recent studies, brain tumours contribute substantially to cancer-related deaths worldwide, highlighting the urgent need for accurate and early diagnosis methods (Xie, 2024). Early detection is critical as tumour progression can rapidly impact surrounding tissues, making prompt intervention essential for improving patient outcomes.

Magnetic Resonance Imaging (MRI) has become the most widely used diagnostic tool for brain tumour detection due to its ability to provide detailed, non-invasive images of brain structures (Ucuzal et al., 2019). MRI scans offer high-resolution visualisation of tumour size, shape, and location, enabling doctors to distinguish between different tumour types such as glioma, meningioma, and pituitary tumours (Ucuzal et al., 2019). However, manual interpretation of MRI images is time-consuming and prone to human error, especially when dealing with large datasets or subtle tumour variations (Verghese et al., 2025). This limitation has driven increasing interest in automated diagnostic systems.

Machine learning (ML) and deep learning (DL) techniques have emerged as powerful tools for medical image analysis. These methods can automatically extract relevant features from MRI data and classify tumours with high accuracy, reducing reliance on manual analysis(Verghese et al., 2025). In particular, Convolutional Neural Networks (CNNs) have performed well in image classification tasks by learning complex patterns directly from data without requiring manual feature engineering (Vidhya et al., 2024). Despite these advancements, challenges remain, including high computational costs, data dependency, and variability in tumour characteristics.

This project aims to explore and evaluate multiple machine learning algorithms for brain tumour classification using MRI data. By comparing different approaches, the study seeks to identify models that balance accuracy, efficiency, and practicality for real-world medical applications.

Literature Review

Machine Learning in Brain Tumour Detection

Machine learning has been widely applied in brain tumour classification, primarily using MRI datasets. Standard ML approaches rely on feature extractions such as texture, intensity, and shape analysis before applying classifiers like Support Vector Machines (SVM), k-Nearest Neighbours (KNN), and Random Forests (RF) (Vidhya et al., 2024). These methods have shown reasonable efficiency; however, they depend heavily on manual input and features designed and engineered by human input, which could fail to capture complex tumour characteristics that humans cannot identify(Verghese et al., 2025).

New research highlights the combination of feature extraction with classification algorithms. For example, shape-based feature extraction methods have been used to distinguish between malignant and benign tumours, improving classification accuracy (Gore et al., 2025). In the same way, studies using logistic regression and random forest models show how feature engineering and optimisation techniques can enhance classification performance while reducing computing costs (Xie, 2024).

Deep Learning Approaches

Deep learning has revolutionised medical image analysis by eliminating the need for manual feature extraction. CNNs, in particular, automatically learn hierarchical features from MRI images, enabling more accurate tumour classification (Vidhya et al., 2024). Studies show that CNN-based models outperform traditional ML approaches in both accuracy and efficiency due to their ability to capture complex spatial patterns in images (Vidhya et al., 2024).

Advanced deep learning techniques, including transfer learning and multi-path networks, have achieved classification accuracies exceeding 98% in some cases (Xie, 2024). However, these methods are immensely complex, requiring significant computing power and large datasets, which limits their practical implementation in environments with resource constraints (Xie, 2024). In order to address this, studies propose using both approaches together, utilising deep learning feature extraction with traditional ML classifiers to balance performance and efficiency.

Image Processing and Feature Engineering

Image preprocessing plays an important role in ensuring models operate at their best capacity. Techniques such as noise removal, contrast enhancement, and segmentation help isolate tumour regions and enhance relevant features (Somasundaram and Gobinath, 2019). Segmentation, in particular, is essential for identifying tumour boundaries and allowing for precise feature extraction.

Feature engineering approaches, including texture and shape analysis, are still important in standard machine learning algorithms. For instance, features like tumour area, perimeter, and compactness can provide valuable insights into tumour characteristics and improve classification (Gore et al., 2025). However, these methods require specialised skills and may not generalise well across different datasets.

Challenges and Limitations

Despite the advancements, many challenges persist in brain tumour classification. One issue is the variability in tumour appearance, including differences in size, shape, and location, which complicates classification tasks (Somasundaram and Gobinath, 2019). Additionally, datasets can be imbalanced and small, leading to limited training data, which can result in overfitting, making models unsuitable for broader use.

Another limitation is the reliance on computationally intensive models, particularly for deep learning approaches. While these models achieve a high accuracy, they often require powerful hardware and large datasets, making them less accessible in smaller settings with limited resources (Xie, 2024). Additionally, manual segmentation and feature extraction in traditional methods are time-consuming and prone to error (Verghese et al., 2025).

Synthesis

The literature indicates that traditional machine learning as well as deep learning approaches are effective for tumour classification, with each possessing advantages and limitations. Traditional ML models are computationally efficient and interpretable, but rely on manual human input for feature extraction. On the other hand, deep learning models offer

superior accuracy by automatically learning features but require significant computational resources and large datasets.

Recent trends suggest that hybrid approaches, combining feature engineering with deep learning or ensemble methods, have presented a promising direction for improving classification performance. Additionally, advancements in computational power and an increasing availability of MRI datasets are expected to further enhance the effectiveness of automated tumour detection systems.

This paper adds to existing research by evaluating multiple machine learning algorithms, aiming to identify approaches that achieve high accuracy while maintaining practicality for real-world medical applications.

Methodology

Dataset and Data Preparation

The dataset used in this study is a *brain tumour classification dataset* from Kaggle consisting of 3,762 instances and 15 attributes (Bohaju, 2020). Each instance represents extracted statistical features from brain MRI images, including measures such as mean intensity, variance, standard deviation, entropy, skewness, kurtosis, contrast, energy, homogeneity, and correlation. The target variable is labelled "*Class"* where values indicate if there is a tumour (=1) or if there is not a tumour (=0). The dataset also contained an *"Image"* column, which was removed during preprocessing as it did not contribute to the model. The data set was split into training and test data being 80 and 20 per cent of the sample, respectively. This was done using a fixed random state to ensure it could be reproduced. All the inputs were also standardised to ensure fairness.

Classification Algorithms

Three supervised machine learning algorithms were utilised based on their common usage in medical image classification and their frequent appearance in the selected articles.

- K-Nearest Neighbour (KNN): A distance-based classifier that assigns labels based on the majority class among the nearest neighbours. In this study, the number of neighbours was set to 5, and scaled data was used to ensure optimal performance.
- Decision Tree (DT): A rule-based classifier that splits the dataset into branches based on features, allowing for clear classification.
- Random Forest (RF): An ensemble method that utilises multiple decision trees to improve the accuracy of classification and reduce overfitting. For this paper, the model used 100 estimators.

Each model was trained with 80% of the data set and then tested with the remaining 20%.

Evaluation Metrics

Model performance was evaluated using accuracy and confusion matrices. Accuracy is measured by the proportion of correctly classified instances, providing an overall performance indicator. Additionally, a bar chart comparison was used to visualise and compare model accuracy.

Confusion matrices were generated for each model to analyse:

- True Positives (correct tumour detection)
- True Negatives (correct non-tumour detection)
- False Positives (incorrect tumour detection)
- False Negatives (incorrect tumour non-detection)

The test data was 723 entries, and each model registered the following in their confusion matrix. KNN registered 324 True Positives, 417 True Negatives, 2 False Positives and 10 False Negatives. Decision Tree registered, 322 True Positives, 412 True Negatives, 7 False Positives and 12 False Negatives. Random Forest registered, 325 True Positives, 416 True Negatives, 3 False Positives and 9 False Negatives

Results

Model Performance

The results obtained from the three models achieve the following accuracy scores:

- K-Nearest Neighbour (KNN): Accuracy = 0.9841
- Decision Tree (DT): Accuracy = 0.9748
- Random Forest (RF): Accuracy = 0.9841

Both KNN and Random Forest achieved the highest accuracy, indicating strong classification performance on the dataset.

Confusion Matrix Analysis

The confusion matrices (Appendix B) provide further insight into model predictions :

- KNN: Correctly classified the majority of cases, with only a small number of misclassifications.
- Decision Tree: Demonstrated slightly lower performance, with more classification errors compared to the other models.
- Random Forest: Produced highly accurate predictions with minimal errors, showing strong generalisation capability.

Across all models, the number of false positives and false negatives remained low, indicating reliable classification performance.

Comparative analysis

The bar chart (Appendix B) shows the comparison between the accuracy of all three ML algorithms. While all models achieved above 97% accuracy, KNN and Random Forest were slightly better. With KNN and Random Forest both achieving the same high level of accuracy, it is clear that the models are capable of identifying patterns in the data set. Random Forest may be slightly better as its ensemble nature will help avoid overfitting.

Discussion

The results of this study demonstrate that machine learning models can achieve very high accuracy in brain tumour classification, with all models reaching 97% accuracy and both KNN and Random Forest achieving 98.4%. This shows that the dataset contains highly informative features that allow for effective separation between tumour and non-tumour cases.

A key reason for this strong performance is the nature of the features used. The dataset includes statistical and texture-based attributes such as mean, variance, entropy, contrast, and homogeneity, which capture important characteristics of MRI images. These types of features are commonly used in medical image analysis and have been shown in the literature to have a strong predictive power for tumour classification. The ability to represent intensity variation and structural patterns makes them particularly useful for distinguishing between normal and abnormal tissue.

The three models selected in this study represent different machine learning approaches: KNN is distance-based, Decision Tree is rule-based, and Random Forest is an ensemble method. This combination allows for a balanced comparison across model types. Random Forest achieved the highest overall performance, which is consistent with expectations. By combining multiple decision trees, it reduces overfitting and improves generalisation, making it more thorough than a single decision tree. The slightly lower performance of the Decision Tree model may be due to its tendency to overfit when not tuned. KNN also performed strongly, which highlights the importance of preprocessing, as KNN relies on distance calculations. To process, feature scaling was applied using StandardScaler to ensure all features contributed equally. Without scaling, features with larger values would have dominated the model. The high accuracy achieved by KNN suggests that the scaling process was effective and that the dataset is well-suited to distance-based classification.

These findings are consistent with previous research, where machine learning models typically achieve accuracy levels above 90% in tumour classification tasks. The strong performance of Random Forest in particular aligns with studies showing that ensemble methods often outperform simpler models due to their ability to capture complex relationships in data.

Despite these positive results, there are several limitations. Only three basic models were tested, and no hyperparameter tuning was applied, meaning performance could potentially be improved further. In addition, accuracy was the only evaluation metric used. While accuracy provides a useful overview, it does not fully reflect the importance of different types of errors. In medical applications, false negatives are especially critical, as failing to detect a tumour can have serious consequences. Future work should therefore include additional metrics such as precision, recall, and F1-score.

In summary, these results highlight the potential of machine learning in supporting medical diagnosis. Such models could assist clinicians by providing fast and consistent predictions, reducing workload and improving decision-making. However, given the limitations, further testing on larger and more diverse datasets would be necessary before real-world implementation. Overall, this study shows that classical machine learning techniques, when combined with appropriate preprocessing and meaningful features, can provide accurate and reliable results for brain tumour classification.

Conclusion

This study analysed the performance of three machine learning algorithms, K-Nearest Neighbour, Decision Tree and Random Forest, for their ability to detect brain tumours using data gathered from an MRI machine. Results indicated that these models are capable of predicting with high levels of accuracy whether a tumour is present.

Random Forest proved to be the most accurate, while its accuracy score was tied with KNN. Given its ensemble format, it can reduce overfitting, making it more practical across a wider range of uses. These findings align with existing research that analyses the ability of machine learning to support the medical field.

The study does have limitations, including the use of a single clean dataset, only using a small group of algorithms and using accuracy as the sole evaluation metric. Future research should focus on a broader dirty dataset, while also potentially exploring other ML algorithms. Overall, this paper displays the promising impact that ML algorithms can have in the medical field to assist doctors with diagnosis, though there will have to be more tests done before it can reliably be used in a real-world application.

Appendix A

Reference List

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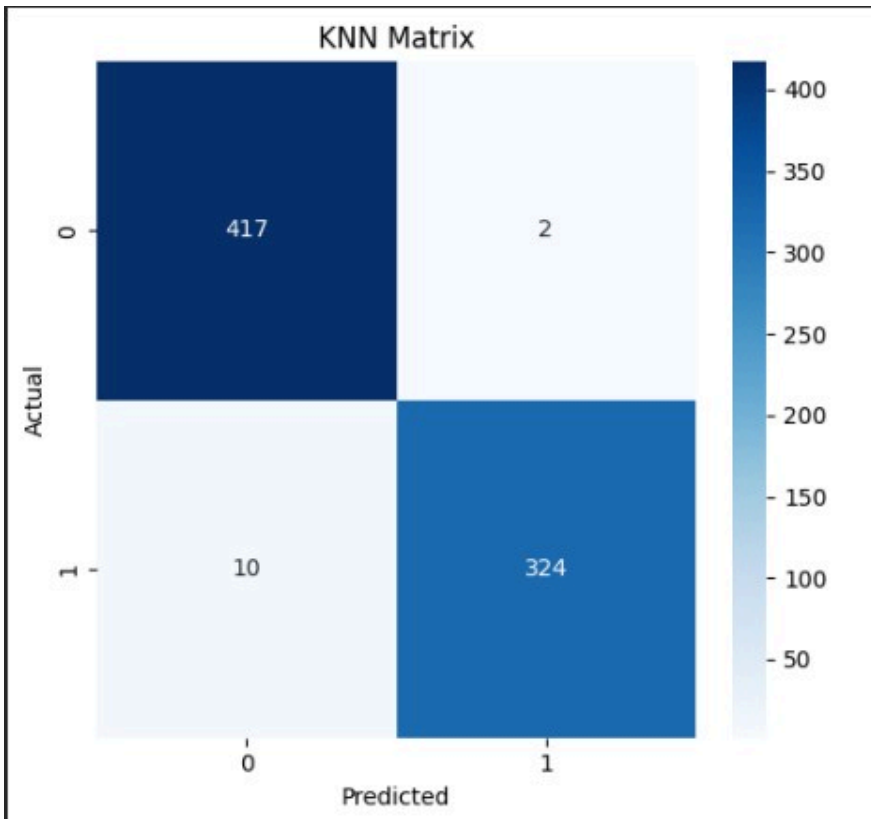
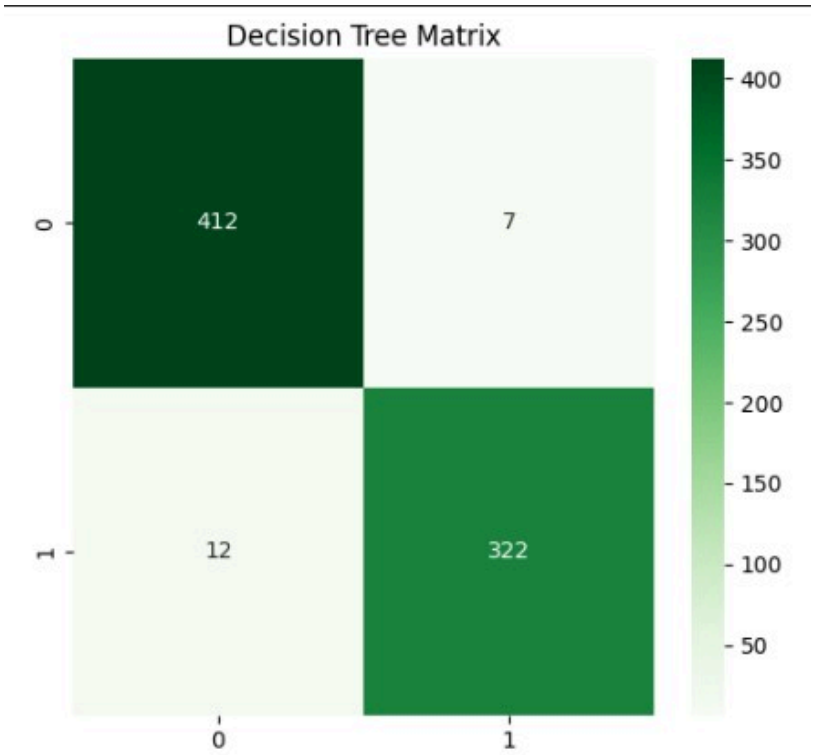
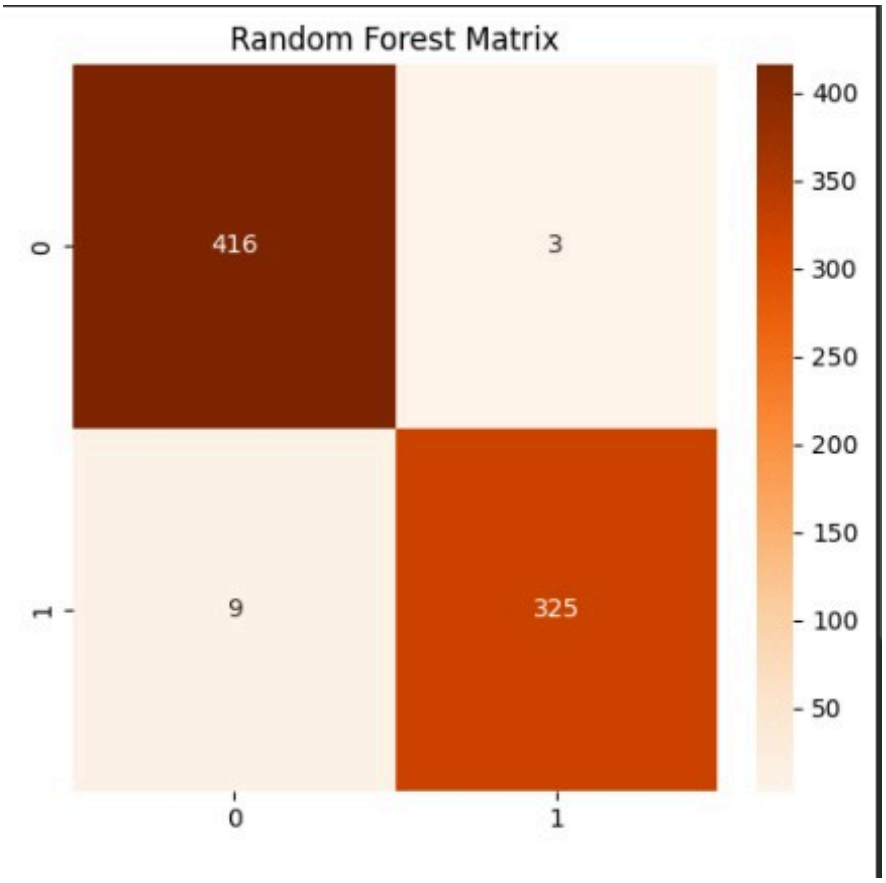
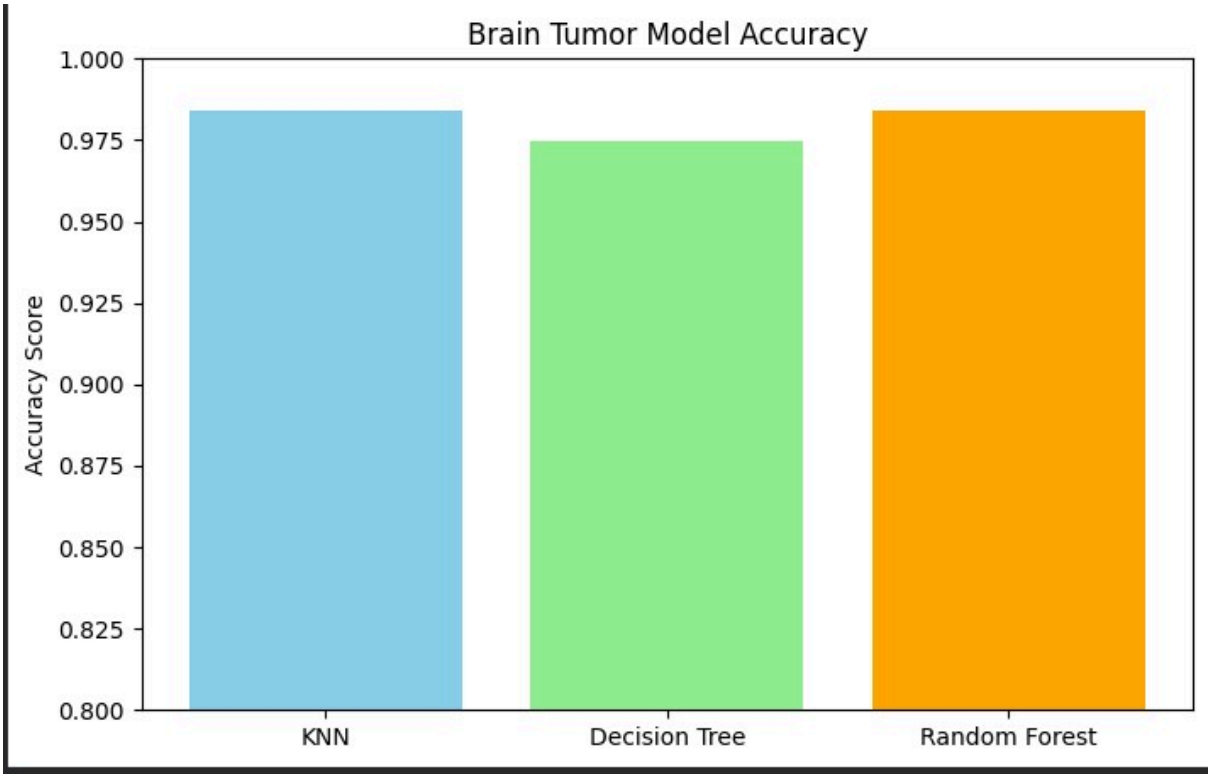
APPENDIX B

Evaluation Graphics










```
Final Results Table:
           Model  Accuracy
0            KNN  0.984064
1  Decision Tree  0.974768
2  Random Forest  0.984064
```

APPENDIX C

Python Code

```python
# We begin by importing everything we will need
# we first import our libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

#we then import the data cleaners and displayers
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score, confusion_matrix

#We also import the ML alrogithms we will be using
from sklearn.neighbors import KNeighborsClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier

# Loading the new brain tumor dataset
df = pd.read_csv('Brain Tumor.csv')

# Inspection steps from your tutorials
print("Dataset Head:")
print(df.head())
print("\nDataset Info:")
print(df.info())

# Preprocessing the data
# Removing the 'Image' column as it's just a filename, not a feature we can access
X = df.drop(columns=['Image', 'Class'])

# We target the 'Class' column (0 or 1)
y = df['Class']

# Split data into training (80%) and testing (20%)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Standardising our features for KNN performance
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

```python
# Traning our three models
# 1. Trainig the K-Nearest Neighbors model
knn = KNeighborsClassifier(n_neighbors=5)
knn.fit(X_train_scaled, y_train)
knn_preds = knn.predict(X_test_scaled)

# 2.Trainig the  Decision Tree model
dt = DecisionTreeClassifier(random_state=42)
dt.fit(X_train, y_train)
dt_preds = dt.predict(X_test)

# 3.  Trainig the Random Forest model
rf = RandomForestClassifier(n_estimators=100, random_state=42)
rf.fit(X_train, y_train)
rf_preds = rf.predict(X_test)

# Creating our graphics
# Graphic 1. Confusion Matrices
plt.figure(figsize=(15, 5))

# Making our KNN Matrix
plt.subplot(1, 3, 1)
sns.heatmap(confusion_matrix(y_test, knn_preds), annot=True, fmt='d', cmap='Blues')
plt.title('KNN Matrix')
plt.xlabel('Predicted')
plt.ylabel('Actual')

# Making our Decision Tree Matrix
plt.subplot(1, 3, 2)
sns.heatmap(confusion_matrix(y_test, dt_preds), annot=True, fmt='d', cmap='Greens')
plt.title('Decision Tree Matrix')

# Making our Random Forest Matrix
plt.subplot(1, 3, 3)
sns.heatmap(confusion_matrix(y_test, rf_preds), annot=True, fmt='d', cmap='Oranges')
plt.title('Random Forest Matrix')

plt.tight_layout()
plt.show()
```

```python
# Graphic 2. Comparison Bar Graph
# collecting our accuracy scores
model_names = ['KNN', 'Decision Tree', 'Random Forest']
scores = [
    accuracy_score(y_test, knn_preds),
    accuracy_score(y_test, dt_preds),
    accuracy_score(y_test, rf_preds)
]
#creating the bar chart that shows the accuracy scores ahansy one aother
plt.figure(figsize=(8, 5))
plt.bar(model_names, scores, color=['skyblue', 'lightgreen', 'orange'])
plt.title('Brain Tumor Model Accuracy')
plt.ylabel('Accuracy Score')
plt.ylim(0.8, 1.0)
plt.show()

# Graphic 3. Summary Table
results_df = pd.DataFrame({
    'Model': model_names,
    'Accuracy': scores
})
#printing a header for our accuracy scores display
print("\nFinal Results Table:")
print(results_df)
```

APPENDIX D

Generative AI Declaration Statement

Declaration Statement

Please tick the appropriate box relating to your use of Gen AI tools for the module Business Strategy (BAA1019)

A. ☐ I did not use any Gen AI tools (including for editing and proofreading) in the completion of this assignment

B.☑ I did use Gen AI tools in the completion of this assignment

Signature: Aidan Mistry

Date: April 21st, 2026

Appendix E

Reporting of Gen AI Tool Use in Assignments

Tools: NotebookLM, used between 04/03/2026 - 17/04/2026	Tools: Google Gemini, used between 04/03/2026 - 17/04/2026	Tools: ChatGPT, used between 04/03/2026 - 17/04/2026
Outputs: NotebookLM was used to draw comparisons between my selected sources and analyse them for relevance.	Outputs: I utilised Google Gemini to help refine my code	Outputs: As I reached the maximum word count, ChatGPT was utilised to identify points that were repeated excessively in my writings
Prompts example: Compare paper X to paper Y in terms of which ML algorithms are used	Prompts example: “What is the cause of this error?” ‘Where in the code is overly complex”	Prompts: Read this paper and identify themes or points that are stated too many times, as well as any overly long wordy sentences
Critical Reflection: For this assignment, I utilised generative AI to assist in technical elements like coding, to assist with my research and to assist with my paper structuring. At no point did AI write anything directly for this paper.		